

Darwin Liao

Software Developer,
M.Sc. Computer Science

Details

Gilford, Ontario
Canada

Links

<https://darwinliao.com/>
<https://github.com/Sherchi>
<https://www.linkedin.com/in/darwin-liao-bb4964155/>

Skills

Python
TypeScript
JavaScript
SQL
React
Next.js
HTML
CSS
Git
Jest
Jira
EC2
Wireshark
TensorFlow
scikit-learn
NumPy
Machine Learning
Anomaly Detection
Feature Engineering
Data Preprocessing
Packet Analysis

Description

Recent M.Sc. Computer Science graduate and software developer with experience in cybersecurity research, machine learning, anomaly detection, and full-stack web development. My work combines Python, TensorFlow, scikit-learn, React, and Next.js to build practical systems. Currently interested in building interactive webpages for hosting events for small communities.

Employment History

Fullstack Developer, Intellijoint Surgical, Kitchener

April 2023 – August 2024

- Collaborated with a cross-functional team of three developers and two QA testers to implement, test, and deliver features.
- Contributed to a surgical pre-planning web application built with a React frontend and Java Spring backend.
- Developed React components and user-facing features, including text inputs, context menus, read-only workflows, and keyboard shortcuts.
- Designed and implemented permission-based case sharing, allowing users to grant others different levels of access to surgical cases.
- Helped build concurrent-access functionality that enabled multiple users to access the same surgical case simultaneously.
- Created a reusable API rate and request size-limiting framework using Java Spring, protecting current and future endpoints against abuse and resource exhaustion.
- Wrote unit tests with Jest and Spring Test for all implemented features and tickets.

Software Validation R&D, General Electric Grid Solutions, Markham

April 2019 – August 2019

- Developed and expanded an automated firmware-testing suite for substation protection relays, validating communication and security functionality across multiple relay models.
- Created relay-specific test arrays and execution workflows to automate firmware tests according to relay type and supported communication protocols.
- Enhanced a universal automated testing and report-generation application, reducing the time, manual effort, and resources required for recurring firmware-validation procedures.
- Tested relay communications within SCADA environments using IEC-61850, Modbus, and TCP/IP.
- Ported, updated, and adapted open-source libraries from C#, C++, and Java to Visual Basic, improving compatibility with the existing codebase and supported relay models.

Software Developer, Aerospace Metal Finishing, Schomberg

April 2018 – August 2018

- Worked with process engineers to develop software to automate reoccurring tasks.
- Gathered and summarized statistics involving different metal finishing jobs to identify slowdowns and issues during different processes.
- Developed a process-tracking application that aggregated the time required for each stage of metal-finishing jobs, helping process engineers identify difficult steps, recurring bottlenecks, and opportunities for workflow improvement.

Education

M.Sc. Computer Science – Dist. Systems, Western University, London

September 2024 – August 2026

Bachelors Computer Science, Western University, London

Sept 2019 – April 2024

Projects

Project Lumi – Eternal Return Game Event Hub

<https://github.com/llyfue/project-lumi>

Tri-Yearly event that is created and hosted by members of the Eternal Return Community. Responsible for the website creation for the events which change theme and premise every occurrence. The website hosts a leaderboard for event participants, and a bounty board where users can submit goals for the participants to complete, and a video section where the participants can submit videos of them completing the goal so viewers can watch.

Project Lumi – Eternal Return Fantasy League

<https://github.com/llyfue/project-lumi>

Website that hosts the fantasy league for the North American Eternal Return competitive circuit. Users create fantasy teams using the most recent players in the professional circuit. Players obtain points based on performance for the week's games, and users score against each other.

Research & Publications

2025 – 9th Cyber Security in Networking Conference (CSNet)

Abu Dhabi, 10.1109/CSNet67572.2025.11288195

OTArmor: Securing Automotive Over-the-Air Updates Against Malware Using Generative Modeling

2026 - Master's Thesis <https://uwo.scholaris.ca/items/4ed13b15-40a4-4b97-8a9e-3e3bf9f5b9b3>

Lightweight Static File Analysis for Pre-Installation Anomaly Screening in Automotive OTA Updates